

OSHAWA, ON, Canada

WMO: 716970

Lat: 43.920N

Lon: 78.893W

Elev: 140

StdP: 99.66

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.6	(c) -17.5	(d) -25.7	(e) 0.4	(f) -20.0	(g) -22.3	(h) 0.5	(i) -16.6	(j) 11.7	(k) -4.8	(l) 10.7	(m) -7.5	(n) 3.7	(o) 0	(p) 0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 29.8	(d) 21.7	(e) 28.2	(f) 21.0	(g) 26.9	(h) 20.2	(i) 23.4	(j) 27.5	(k) 22.5	(l) 26.2	(m) 21.7	(n) 25.1	(o) 4.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.0	25.5	21.2	16.1	24.7	20.4	15.3	23.9	70.4	27.7	66.8	26.3	63.9	25.1	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	9.0	8.0	-24.7	32.9	2.7	2.2	-26.7	34.5	-28.3	35.8	-29.8	37.1	-31.8	38.7
			-25.0	24.9	2.8	1.2	-27.0	25.8	-28.6	26.5	-30.2	27.1	-32.2	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-5.7	-5.2	-0.7	5.6	13.5	18.1	21.4	20.2	16.4	9.8	2.9	-1.8
	DBStd	10.61	6.31	6.29	5.81	4.14	4.32	3.17	2.74	2.64	3.95	4.31	4.53	5.30
	HDD10.0	2043	485	427	335	141	16	1	0	0	1	58	214	366
	HDD18.3	4028	743	660	591	381	161	42	6	12	84	264	462	624
	CDD10.0	1298	0	0	3	10	123	244	353	316	193	53	3	0
	CDD18.3	242	0	0	0	0	9	35	101	69	27	1	0	0
	CDH23.3	1711	0	0	0	1	68	227	787	443	183	3	0	0
	CDH26.7	338	0	0	0	0	7	40	196	64	32	0	0	0
Wind	WSAvg	3.7	4.2	4.2	4.3	4.4	3.5	3.3	3.0	2.9	3.0	3.5	3.7	3.8
Precipitation	PrecAvg	882	66	53	60	79	77	76	82	76	81	83	72	72
	PrecMax	1039	127	110	124	126	143	155	166	124	154	167	112	147
	PrecMin	753	34	24	28	28	24	19	36	28	45	27	30	42
	PrecStd	89	26	20	24	30	33	38	36	22	27	35	22	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	10.5	19.0	20.9	27.3	30.1	32.3	30.6	29.6	23.4	16.3	12.1
		MCWB	7.8	7.1	15.2	13.7	19.4	20.5	22.9	22.5	22.9	17.7	11.9	9.9
	2%	DB	6.3	7.4	12.8	17.5	25.3	27.7	30.4	28.3	27.3	20.8	13.6	8.9
		MCWB	5.0	5.3	8.7	11.1	18.2	20.2	22.2	21.5	21.7	16.7	10.6	7.4
	5%	DB	4.6	5.1	9.6	14.9	23.2	25.9	28.8	27.0	25.4	18.7	12.0	6.9
		MCWB	3.5	3.5	7.2	10.1	16.5	18.6	21.0	20.6	20.1	15.1	9.5	5.9
	10%	DB	2.8	3.0	7.2	12.7	21.3	24.2	27.2	25.9	23.6	16.8	10.2	5.1
		MCWB	1.8	1.5	4.7	8.4	15.4	18.0	20.1	20.1	19.4	14.3	8.0	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	7.7	15.1	15.8	21.0	23.0	25.1	24.0	23.8	20.0	13.0	10.5
		MCDB	9.0	8.5	19.1	19.4	26.0	27.9	30.6	27.8	27.8	21.0	14.9	11.5
	2%	WB	5.4	5.7	10.2	12.9	19.3	21.5	23.6	23.1	22.5	17.4	11.2	7.7
		MCDB	6.1	7.3	12.0	14.9	23.6	25.4	28.4	26.2	26.1	19.6	13.0	8.5
	5%	WB	3.5	3.6	7.5	11.1	17.8	20.4	22.5	22.2	21.4	15.8	9.7	6.0
		MCDB	4.4	4.9	9.4	14.1	22.1	23.8	26.7	25.3	24.3	17.9	11.5	7.1
	10%	WB	1.9	1.6	5.0	9.1	16.3	19.3	21.6	21.4	20.1	14.5	8.2	4.0
		MCDB	2.8	2.9	6.8	11.9	19.7	22.6	25.3	24.4	22.6	16.2	10.1	5.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	8.5	9.4	10.2	9.8	10.4	9.8	10.0	8.7	8.1	6.5
		MCDBR	7.8	9.4	11.3	13.1	12.7	11.7	12.2	11.0	11.3	11.0	10.7	7.5
		MCWBR	7.5	8.1	8.2	8.7	6.8	6.0	5.9	5.7	6.3	7.5	8.6	7.1
	5% WB	MCDBR	8.2	9.2	10.4	11.0	10.8	9.9	10.4	8.9	10.2	9.3	9.5	7.2
		MCWBR	8.1	8.3	8.2	8.5	6.5	6.1	5.5	5.2	6.2	7.1	8.2	7.2
Clear-Sky Solar Irradiance	taub	0.303	0.311	0.329	0.372	0.402	0.430	0.427	0.415	0.379	0.361	0.317	0.292	
	taud	2.351	2.335	2.406	2.334	2.299	2.248	2.270	2.292	2.378	2.410	2.490	2.448	
	Ebn at Noon	817	879	911	892	871	844	841	839	844	811	803	803	
	Edh at Noon	79	96	102	120	128	135	131	124	105	89	70	66	
All-Sky Solar Radiation	RadAvg	1.34	2.17	3.46	4.45	5.43	5.85	5.92	5.19	4.13	2.45	1.51	1.08	
	RadStd	0.13	0.20	0.34	0.47	0.58	0.45	0.43	0.26	0.36	0.23	0.19	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page